

1 The Hamiltonian Operator

1.1 Introduction to the Hamiltonian

In quantum computing, we often focus on the "what"—what gate to apply or what algorithm to run. But there is a more fundamental "how" happening behind the scenes. Rather than just diving straight into the definition, we need to realise the need for it: to understand how a quantum computer functions on a physical level, we must understand the Hamiltonian.

1.2 The Process of Evolution

When we look at a quantum circuit, we see gates such as the Hadamard (H) gate or the X gate acting on qubits. In circuit diagrams, it looks as if the qubit's state suddenly "jumps" from one state to another.

For example, consider a single qubit initially in state $|0\rangle$:

$$|0\rangle \xrightarrow{H} \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

The circuit diagram shows only the starting state and the final state after the gate. It does not show what happens in between.

In reality, quantum systems cannot instantly jump from one state to another. According to quantum mechanics, the state of a closed quantum system changes continuously over time. This continuous evolution is described by the Schrödinger equation.

A useful analogy is driving a car from one city to another. A circuit diagram is like a map that only marks the starting city and the destination. It does not show the actual road taken. The quantum gate represents the destination, while the Hamiltonian acts like the engine and steering system that guides the car along a valid route.

For instance, an X gate changes $|0\rangle$ to $|1\rangle$:

$$|0\rangle \xrightarrow{X} |1\rangle$$

The circuit suggests an immediate flip, but physically the qubit rotates smoothly through intermediate states before reaching $|1\rangle$. The Hamiltonian governs this rotation and ensures that the evolution follows the laws of quantum mechanics.

Another way to think about it is a movie versus a photograph. A quantum gate in a circuit diagram is like seeing only the first and last frames of a movie. The Schrödinger equation

provides all the missing frames, showing how the quantum state evolves moment by moment. The Hamiltonian determines what those frames look like.

1.3 Formal Definition

The Hamiltonian is one of the most important concepts in quantum mechanics. It is often defined as the operator that represents the total energy of a quantum system.

In mathematical terms, the Hamiltonian is a matrix that contains information about the possible energy levels of the system and the quantum states associated with those energy levels.

To understand this, imagine a building with several floors, each floor represents a possible energy level of the system (an energy eigenvalue). The rooms on that floor represent the corresponding energy eigenstates. If a particle is in one of these special states, it stays on the same floor over time; it does not move to another energy level on its own. The state only acquires a phase, which is like the room's clock ticking while everything else remains unchanged.

Optional Background: Terms such as *energy eigenstates* and *energy eigenvalues* come from quantum mechanics and are not required for understanding the basic role of the Hamiltonian in quantum computing. Interested readers can explore them in [Quantum Computation and Quantum Information \(Chapter 2\)](#) after becoming comfortable with qubits and quantum gates.

1.4 Practical Significance

At first, the Hamiltonian may seem unimportant if you're only working with quantum gates like X, Z, or Hadamard. However, it plays a crucial role because it connects a quantum algorithm to the actual hardware that runs it.

For now, the most important takeaway is that the Hamiltonian is the underlying engine that drives the physical implementation of quantum operations over time.